

NOR-TL: NORMAL-REFERENCE TEMPORAL LEARNING WITH DECOUPLED DETECTION AND TEMPORAL SUPERVISION FOR PAROXYSMAL AF EPISODE LOCALIZATION

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ABSTRACT

While single-lead electrocardiogram (ECG) classification is mature, paroxysmal atrial fibrillation (PAF) localization is hindered by the scarcity of dense rhythm annotations. Such limited annotated cohorts restrict the population diversity available for model training. To break this bottleneck, we propose NOR-TL, a framework that unifies large-scale coarse pre-training with fine-grained interval supervision. To accommodate imprecise coarse recording labels, NOR-TL leverages a pre-trained encoder to extract transferable normal-reference features and an adaptive stream for AF-specific modeling. It adopts a multiple instance learning (MIL) temporal predictor to decouple supervision tasks. Weak coarse labels guide pooled detection learning, and dense interval annotations constrain sample-wise temporal supervision. Evaluated on AFDB, LTAfDB, and CPSC2021, NOR-TL achieves top performance (97.98% temporal AUROC, 0.838 MCC) on CPSC2021, demonstrating the efficacy of combining complementary annotation granularities.

Index Terms— Paroxysmal atrial fibrillation, Single-lead ECG, Deep learning, Multi-granularity supervision

1. INTRODUCTION

Automated arrhythmia detection from short single-lead electrocardiogram (ECG) recordings has achieved strong diagnostic performance [1, 2]. In portable and home-monitoring scenarios, however, paroxysmal atrial fibrillation (PAF) monitoring requires more than identifying whether AF is present in a recording. This requires solving two tightly coupled tasks: *window-level AF recognition*, which determines whether an ECG segment contains AF, and *time-point localization*, which identifies the temporal extent of AF episodes and their transitions to surrounding non-AF rhythm. The latter is particularly important for estimating AF burden [3, 4]. Yet densely annotated datasets suitable for temporal localization remain limited in patient diversity; AFDB, for example, provides only 23 waveform records [5]. Splitting long recordings into many windows increases sample quantity without increasing patient diversity, while ECG models are known to be sensitive to cross-database distribution shifts [6].

Large-scale public ECG datasets provide a complementary source of population diversity for representation learning [7, 8, 9]. Pretraining on such data can establish general ECG feature-extraction capabilities across diverse patient populations and recording conditions [10, 11, 12]. However, labels for these large-scale

datasets are typically available only at the recording level. Applying a positive recording label to every time point would incorrectly supervise non-AF portions of mixed-rhythm windows. Multiple instance learning [13] provides a way to exploit such coarse labels through pooled evidence without converting them into temporal ground truth. Therefore, effective PAF localization requires both transferable representations learned from large-scale data and supervision mechanisms matched to different annotation granularities.

For these representations to support temporal localization, they must capture both AF and non-AF rhythm characteristics, as PAF episodes alternate with non-AF intervals. Previous work has shown that combining complementary waveform morphology and rhythm information improves robust AF detection [14]. We further posit that AF-specific features can benefit from an explicit normal-reference context. Accordingly, we use a dual-encoder formulation in which a normal-pretrained encoder remains frozen to preserve a stable normal-trained mapping, while a trainable AF encoder adapts to the downstream task. Both encoders process the same ECG, allowing their temporally aligned features to be fused for local rhythm discrimination.

Here, we propose NOR-TL, a normal-reference temporal learning framework with annotation-specific supervision. A shared temporal predictor supports both *window-level AF recognition* and *time-point localization*: weak recording labels supervise pooled detection, whereas dense interval annotations provide sample-wise temporal supervision. This design enables large-scale recording-labeled data to contribute beyond pretraining while preserving the temporal information available in densely annotated datasets. Our contributions are threefold:

(1) Normal-reference temporal learning. We introduce a dual-stream representation that combines a frozen normal-pretrained encoder with an adaptive AF encoder, providing stable normal-reference context together with task-specific AF features.

(2) Annotation-specific supervision across label granularities. We route weak recording labels to pooled window-level detection and dense interval labels to sample-wise temporal localization on a shared predictor, avoiding the direct assignment of coarse labels as temporal targets.

(3) Multi-database evaluation. We evaluate both window-level AF recognition and time-point localization on AFDB, LTAfDB, and CPSC2021 against three locally adapted published methods, with controlled ablations of normal-reference conditioning and supervision routing.

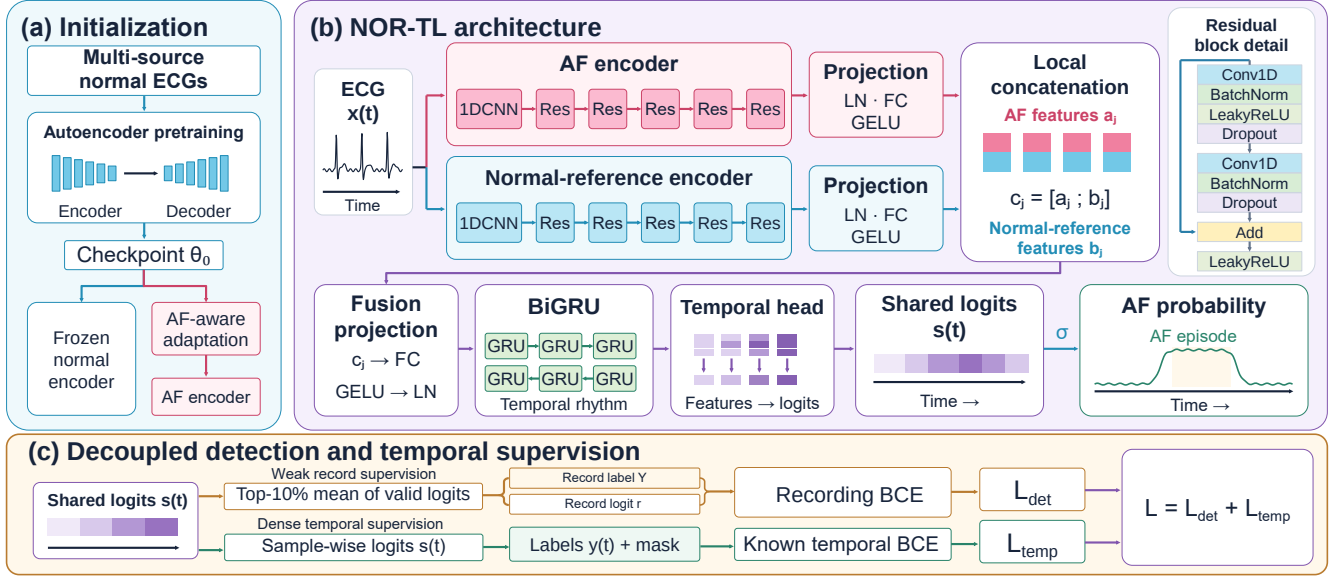


Fig. 1. Overview of NOR-TL. (a) Waveform reconstruction on normal ECGs and AF adaptation learn normal-reference and AF-specific representations. (b) The frozen normal encoder and trainable AF encoder process identical ECG inputs; their features are fused for temporal prediction. (c) Weak recording labels supervise pooled detection, and dense interval labels drive sample-wise localization via the shared predictor.

2. METHOD

2.1. Episode localization and model overview

NOR-TL processes single-channel ECGs in 10-s windows, each represented as $x_i \in \mathbb{R}^{2500}$. Training uses two annotation types: a recording label Y_i indicates AF presence, whereas a temporal label y_{it} identifies AF or non-AF at sample t . Temporal labels are obtained from annotated episode intervals; unknown portions and padding are excluded from their loss. The target follows the AF-family labels, including atrial fibrillation and atrial flutter.

As shown in Fig. 1(b), two encoders extract corresponding local features from the same ECG. Their fused sequence produces temporal AF probabilities p_{it} , which support interval localization. Pooling the underlying logits also gives a recording probability q_i , allowing recording and interval annotations to supervise the shared predictor at their respective resolutions.

2.2. Normal-reference temporal learning

We first pretrain a one-dimensional residual autoencoder on multi-source normal ECGs, using waveform reconstruction to establish its feature-extraction capability (Fig. 1(a)). The pretrained encoder initializes the normal-reference stream. A copy undergoes AF-specific adaptation with AF-positive, normal, and non-AF abnormal examples to initialize the AF stream (Section 3.2). The decoder is used only during pretraining.

Each encoder uses channel widths of 32, 64, 128, 256, 512, and 512, with kernel size five and successive downsampling factors of 2, 2, 2, 2, and 1. Channel reduction and adaptive pooling produce a 156×128 feature sequence. During downstream learning, the normal encoder remains frozen, while the AF encoder is updated with fixed batch-normalization running statistics.

Let E_A and E_N denote the AF and normal encoders. At each position j , independent trainable projections P_A and P_N , each comprising layer normalization, a linear layer, and Gaussian error linear

unit (GELU), produce

$$a_{ij} = P_A(E_A(x_i)_j), \quad b_{ij} = P_N(E_N(x_i)_j), \quad (1)$$

where $a_{ij}, b_{ij} \in \mathbb{R}^{128}$ and $j = 1, \dots, 156$. We concatenate the corresponding features and project them into a shared temporal representation:

$$u_{ij} = \text{LN}(\text{GELU}(W_f[a_{ij}; b_{ij}] + c_f)), \quad (2)$$

where $W_f \in \mathbb{R}^{256 \times 256}$ and $c_f \in \mathbb{R}^{256}$ are the fusion weights and bias. This fusion provides the temporal model with adaptive AF features and normal-trained context at each position.

A single-layer bidirectional gated recurrent unit (BiGRU) with 64 hidden units per direction processes the fused sequence. Layer normalization and a linear head produce local logits, followed by nearest-valid edge extension and linear interpolation to the 2,500-sample input grid. The resulting logits s_{it} yield $p_{it} = \text{sigmoid}(s_{it})$. Valid positions are packed for recurrent processing, while unknown annotation intervals retain their original temporal positions. Recurrent states are reset for each window, providing bidirectional context within that window.

2.3. Decoupled detection and temporal supervision

Let $\mathcal{W}$ denote weakly labeled recordings used for detection supervision and $\mathcal{D}$ denote densely annotated windows with at least one known temporal label. Both sets train the same temporal predictor (Fig. 1(c)). We use pooled logits for detection on $\mathcal{W}$ and sample-wise interval supervision on $\mathcal{D}$; throughout this section, BCE denotes binary cross-entropy computed from logits.

For weakly labeled recordings, top- k pooling aggregates temporal evidence into a recording-level detection score. Let $V_i = \{t : v_{it} = 1\}$ denote valid signal positions, excluding padding, and let K_i index the $k_i = \lceil 0.1|V_i| \rceil$ largest logits in V_i . The recording logit and probability are

$$r_i = \frac{1}{k_i} \sum_{t \in K_i} s_{it}, \quad q_i = \text{sigmoid}(r_i). \quad (3)$$

The pooled logit r_i is supervised by the recording label Y_i :

$$\mathcal{L}_{\text{det}} = \frac{1}{|\mathcal{W}|} \sum_{i \in \mathcal{W}} \text{BCE}(r_i, Y_i). \quad (4)$$

This objective supervises recording-level AF evidence, while the interval objective below supplies the temporal targets.

For a densely annotated window, $m_{it} = 1$ identifies samples that are both valid and temporally annotated. Let $M_i = \sum_t m_{it}$ be their number. Temporal supervision uses the interval labels y_{it} to penalize missed AF time and false-positive extension into non-AF time:

$$\mathcal{L}_{\text{temp}} = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \frac{1}{M_i} \sum_t m_{it} \text{BCE}(s_{it}, y_{it}). \quad (5)$$

The loss is averaged within each window and then across windows, so varying annotation coverage does not change the weight of a window. The two terms are combined with equal weights:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{temp}}. \quad (6)$$

Four-source recordings provide detection supervision, whereas interval-annotated AFDB windows provide temporal supervision.

The gradient directions explain why dense windows receive only interval supervision. For one dense positive window, consider an additional detection loss $\ell_{\text{rec}} = \text{BCE}(r_i, 1)$, and let ℓ_{temp} denote its mean temporal BCE. At a selected, known non-AF sample $t \in K_i$ with $y_{it} = 0$, the gradients are

$$\frac{\partial \ell_{\text{rec}}}{\partial s_{it}} = \frac{q_i - 1}{k_i} < 0, \quad \frac{\partial \ell_{\text{temp}}}{\partial s_{it}} = \frac{p_{it}}{M_i} > 0, \quad (7)$$

with the top- k selection held locally fixed. At a selected non-AF position, the positive detection objective increases the logit, whereas temporal supervision decreases it. To avoid imposing these competing targets on densely annotated windows, NOR-TL applies detection supervision to weakly labeled recordings and temporal supervision to dense windows. Both objectives update the shared trainable components, allowing recording-level data to support localization while interval annotations supervise episode extent.

3. EXPERIMENTS

3.1. Datasets and preprocessing

To make full use of large-scale public ECG datasets, we use CODE15 [7], PTB-XL [8], CPSC2018 [15], and Chapman-Shaoxing [9] for pretraining and recording-level supervision. Lead I emulates the single-lead input setting of wearable and home monitoring. Screening retains usable recordings with AF-family, clean-normal, or non-AF abnormal labels. Table 1 reports the screened cohorts.

AFDB [5] provides dense supervision and held-out testing; the Long-Term AF Database (LTAfDB) [16] and the China Physiological Signal Challenge (CPSC) [17] provide external evaluation. Long recordings are divided into 10-s windows. All windows from the same patient or original recording are kept in a single partition to prevent window-level leakage. Signals are calibrated in millivolts and resampled to 250 Hz.

3.2. Implementation and training

Encoder initialization used 40 epochs of normal ECG pretraining followed by 10 epochs of AF adaptation. NOR-TL was then trained for 10 epochs using AdamW, with learning rates of 10^{-5} for the AF

Table 1. Summary of pretraining and evaluation datasets.

Dataset	Unit	Retained	Train	Val.	Test
CODE15	Recordings	78,962	56,346	10,048	12,568
PTB-XL	Recordings	9,481	7,167	1,243	1,071
CPSC2018	Recordings	5,019	3,792	671	556
Chapman-Shaoxing	Recordings	8,146	6,172	1,068	906
AFDB	Recordings	23	16	2	5
	Windows	84,310	58,892	7,013	18,405
LTAfDB	Recordings	11	—	—	11
	Windows	94,200	—	—	94,200
CPSC2021	Recordings	1,425	—	—	1,425
	Windows	173,537	—	—	173,537

encoder and 10^{-3} for new layers, weight decay 10^{-4} , and gradient clipping at five. Each batch contained 32 weakly and 32 densely labeled examples. Training was performed on an NVIDIA GeForce RTX 5070 Ti GPU. Model checkpoints were selected according to the mean validation recording and temporal AP.

3.3. Results and discussion

3.3.1. Main performance comparisons

We compared NOR-TL with three publicly implemented methods: the waveform CRNN of Wen and Kang [18], SAFER [19], and the three-beat ResNet-Transformer [20]. All three were adapted to single-lead inputs and retrained using common data splits. Additional references include MMA-RNN [21], evaluated on a separate patient-disjoint CPSC2021 split, and published 5-s AFDB classification results for HAN-ECG3 [22].

Table 2 shows that NOR-TL yields the best window-level and time-point Matthews correlation coefficient (MCC) across all three test cohorts. The most prominent performance improvement occurs on CPSC2021, our largest external test set consisting of 1,425 held-out ECG recordings. On this dataset, NOR-TL attains a temporal MCC of 0.838, outperforming SAFER (0.735); sensitivity rises from 73.4% to 85.5% while specificity marginally improves to 96.5%, demonstrating more complete capture of AF episodes with limited false positives on non-AF segments. On AFDB and LTAfDB, NOR-TL also achieves the highest temporal MCCs (0.465 and 0.765, respectively). While its sensitivity is lower than the three-beat baseline, NOR-TL substantially boosts specificity on both datasets, suppressing false positive predictions over non-AF regions. Collectively, the consistently superior temporal MCC across all three cohorts verifies that NOR-TL strikes a favorable balance between AF detection sensitivity and non-AF rejection, demonstrating the efficacy of normal-reference temporal learning coupled with multi-granularity annotation supervision for precise AF episode localization.

Beyond the quantitative comparison, Fig. 2 illustrates AF episode localization in a selected CPSC2021 recording with paroxysmal AF. SAFER produces several false-positive intervals during non-AF rhythm, whereas CRNN and 3-beat R-T intermittently miss portions of the longer AF episode, fragmenting their detections into disconnected intervals. NOR-TL captures both episodes with more continuous coverage and fewer false-positive detections than SAFER. The enlarged ECG segment demonstrates that NOR-TL detects the initial AF onset merely 1.06 s prior to the manually annotated rhythm transition, verifying its capability to localize rhythm

Table 2. AF recognition and temporal labeling performance.

Model	Level	Se	Sp	AUROC	MCC
AFDB					
CRNN [18]	W	75.1	52.3	69.7	0.281
	T	62.8	55.3	69.8	0.180
SAFER [19]	W	37.7	60.1	53.5	-0.022
	T	38.2	60.7	55.1	-0.011
3-beat R-T [20]	W	57.0	91.3	87.0	0.494
	T	86.6	45.7	76.3	0.359
NOR-TL	W	77.2	73.4	87.3	0.503
	T	65.3	81.6	85.3	0.465
LTAfDB					
CRNN [18]	W	74.5	97.1	94.3	0.710
	T	64.4	97.4	94.0	0.632
SAFER [19]	W	77.3	98.9	96.6	0.754
	T	77.6	98.0	96.3	0.754
3-beat R-T [20]	W	67.2	98.5	95.3	0.661
	T	85.2	91.4	95.2	0.761
NOR-TL	W	84.9	98.2	96.4	0.819
	T	79.6	97.3	96.0	0.765
CPSC2021					
CRNN [18]	W	72.7	95.4	94.0	0.720
	T	58.9	96.9	94.7	0.637
SAFER [19]	W	73.6	95.6	94.9	0.731
	T	73.4	95.9	95.4	0.735
3-beat R-T [20]	W	37.6	96.8	91.2	0.457
	T	71.7	87.9	90.5	0.604
NOR-TL	W	92.9	94.1	98.2	0.863
	T	85.5	96.5	98.0	0.838
<i>Independent / published protocols</i>					
MMA-RNN [†] [21]	T	81.6	92.4	93.1	0.735
HAN-ECG3 [‡] [22]	W	90.53	79.54	89.46	–

W: window recognition; T: time-point labeling. [†]MMA-RNN: re-evaluated following its patient-disjoint CPSC2021 setup.

[‡]HAN-ECG3: directly quoted from published 5-s AFDB classification results.

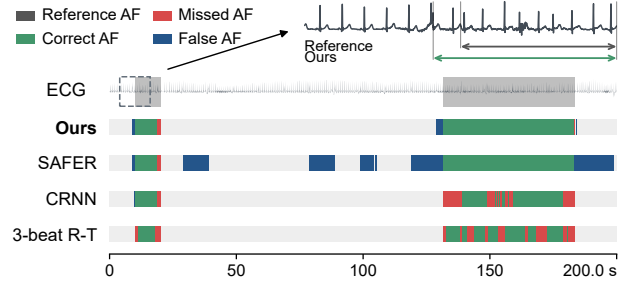
switching points with minimal temporal offset.

3.3.2. Ablation studies

We conducted controlled ablations on AFDB to assess normal-reference conditioning and annotation-specific supervision for time-point localization. All variants follow the same training protocol, with changes restricted to the component under evaluation.

Normal-reference conditioning. Variants B and C demonstrate the contribution of the normal-reference design. Removing the reference encoder (B) reduced temporal MCC from 0.465 to 0.258 and AUROC from 85.3% to 74.7%, with decreases in both sensitivity and specificity. Replacing the normal-pretrained reference with an AF-pretrained encoder (C) increased sensitivity but reduced specificity to 60.0% and MCC to 0.404. The increased AF detection came with substantially more false positives in non-AF intervals. Together, the B/C comparisons highlight the role of normal-reference conditioning in suppressing false-positive AF predictions and improving the balance between AF detection and non-AF rejection.

Annotation-specific supervision. Variants D and E assess the contributions of decoupled supervision and continued use of weak recording labels during downstream learning. Adding recording-

**Fig. 2.** Qualitative comparison of AF episode localization. The inset shows an enlarged ECG segment around an annotated AF onset.**Table 3.** Ablation study on AFDB time-point localization.

Variant	Se	Sp	AUROC	MCC
<i>Normal-reference conditioning</i>				
B: No reference encoder	58.6	67.5	74.7	0.258
C: AF-pretrained reference	79.5	60.0	68.1	0.404
<i>Annotation-specific supervision</i>				
D: + Dense detection BCE	40.4	84.4	78.0	0.268
E: W/o downstream weak BCE	27.7	56.5	51.3	-0.165
A: Full NOR-TL	65.3	81.6	85.3	0.465

level BCE to densely annotated windows (D) increased specificity to 84.4% but reduced sensitivity from 65.3% to 40.4%, yielding lower AUROC (78.0%) and MCC (0.268). Removing downstream weak recording-level supervision (E), while retaining pretrained encoders and dense temporal supervision, severely degraded localization performance: MCC fell to -0.165, sensitivity to 27.7%, and AUROC to 51.3%. Together, these results demonstrate the importance of assigning recording and interval labels to their respective objectives: separate temporal supervision improves AF coverage in dense windows, while continued weak supervision sustains AF discrimination beyond pretraining.

Overall, the full model achieved the highest temporal AUROC and MCC on AFDB, supporting the combined contributions of normal-reference temporal learning and annotation-specific supervision to PAF episode localization.

4. CONCLUSION

We proposed NOR-TL for single-lead PAF episode localization, combining normal-reference temporal learning with decoupled detection and temporal supervision. A normal-pretrained encoder provides stable reference features that are fused with those of an adaptive AF encoder at corresponding temporal positions. Weak recording labels supervise pooled AF detection, while dense interval labels supervise sample-wise localization through a shared predictor. This design allows large-scale recording-labeled data to support localization learning alongside limited interval annotations. Across AFDB, LTAfDB, and CPSC2021, NOR-TL achieved the highest temporal MCC among the adapted baselines, with temporal AUROC and MCC reaching 97.98% and 0.838 on CPSC2021, respectively. Ablations supported the contributions of normal-reference conditioning and decoupled supervision. While the Lead-I ECG settings fit wearable devices, all evaluations in this study are conducted on clinical-grade recordings. Future work will further validate model robustness on real-world wearable ECG data.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used only publicly available, de-identified ECG datasets in accordance with their respective data-use conditions. No new human participant data were collected, and no participants were prospectively recruited for this study. The work involved only secondary analysis of existing de-identified data.

6. ACKNOWLEDGMENT

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